

Project Genesis: Research Orchestrator & 2^4 Factorial Experimentation Infrastructure

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Abstract

Project Genesis has evolved from an isolated artificial life simulator into a scientifically rigorous, high-throughput computational research platform. This technical paper presents the design and implementation of the **Genesis Research Orchestrator**—a system integrating multi-core parallel process management, continuous Phase 9A checkpoint-resume state recovery, failure classification, and automated experimental campaign aggregation. The orchestrator enables systematic 2^4 full factorial experimental screening across arbitrary seed blocks with zero data loss under process interruption.

1 Introduction

Simulating complex socio-ecological interactions and evolutionary dynamics requires running millions of ticks across high-dimensional parameter spaces. To transform raw simulation capabilities into reproducible scientific findings, robust experimental infrastructure is required. The Genesis Research Orchestrator addresses this challenge by providing process clamping, fault-tolerant state recovery, campaign provenance, and automated dataset export.

2 System Architecture

The Genesis platform is structured into three decoupled operational layers:

1. **Layer 3: Research Infrastructure (Orchestrator)**: Manages campaign manifests (`campaign_manifest.json`), concurrency clamping (`BATCH_SIZE = 4`), Phase 9A checkpoint recovery, failure logging, and CSV/JSON summary aggregation.
2. **Layer 2: Research Platform (Analytics & Web Viewer)**: Provides settlement location prediction, spatial heatmaps, trajectory telemetry, and interactive web visualization.
3. **Layer 1: Simulation Engine (Physical & Agent Core)**: Executes the Whittaker biome matrix, priority-flood hydrology, Hebbian learning, and biological drive integration.

3 Study A: 2^4 Full Factorial Experimental Design

Study A evaluates the main effects and higher-order interactions between territorial disputes, spatial dispersal, healing speed, and resource scarcity.

Crossing these four factors yields 16 orthogonal conditions (C_{01} through C_{16}), executed across world seed blocks (e.g., Seed 1720, 1847, 2011) to isolate environmental effects from specific map geographies.

Table 1: Factorial Design Matrix for Study A

Factor	Parameter	Level 0 (−)	Level 1 (+)
F_1	Disputes (<code>disputes_enabled</code>)	False (Social Peace)	True (Hostility)
F_2	Spawn Mode (<code>spawn_mode</code>)	"fixed" (Clusters)	"random_valid" (Dispersed)
F_3	Healing Rate (<code>healing_speed_mult</code>)	1.0 (Standard)	0.5 (Harsh)
F_4	Scarcity (<code>scarcity</code>)	1.0 (Abundant)	3.0 (Depleted)

4 Parallel Concurrency Clamping

Benchmarking on an 8-core / 16-thread processor demonstrated an $8.51\times$ throughput speedup at concurrency $N = 8$. To preserve memory locality and prevent thread thrashing during late-game cognition ($> 10^6$ ticks), active process execution is clamped to `BATCH_SIZE = 4`.

Each experiment executes in an isolated OS console window (`CREATE_NEW_CONSOLE`). The orchestrator monitors active process handles synchronously, launching subsequent batches automatically upon batch completion.

5 Phase 9A Checkpoint & Auto-Resume Engine

To ensure fault tolerance over long execution horizons (10^7 ticks $\approx 27,777$ simulation years), state is saved periodically every 50,000 ticks.

- **Persistent State:** Physiological scalars, drive tensions, Hebbian weights, social relationships, and episodic memory are serialized to `latest_checkpoint.json`.
- **Derived State:** Deterministic 2D NumPy world arrays (elevation, biomes, hydrology) are regenerated from the world seed in ~ 2.5 seconds upon resume.

Upon launcher invocation, existing experiment directories are inspected. Experiments marked as complete are skipped instantly, while interrupted runs load `latest_checkpoint.json` and resume seamlessly from the last saved tick.

6 Conclusion & Future Work

The Genesis Research Orchestrator freezes the core simulation infrastructure into a reliable, reproducible experiment engine. Future work will focus on executing multi-seed campaign blocks and analyzing multi-factorial interactions using linear mixed-effects models.